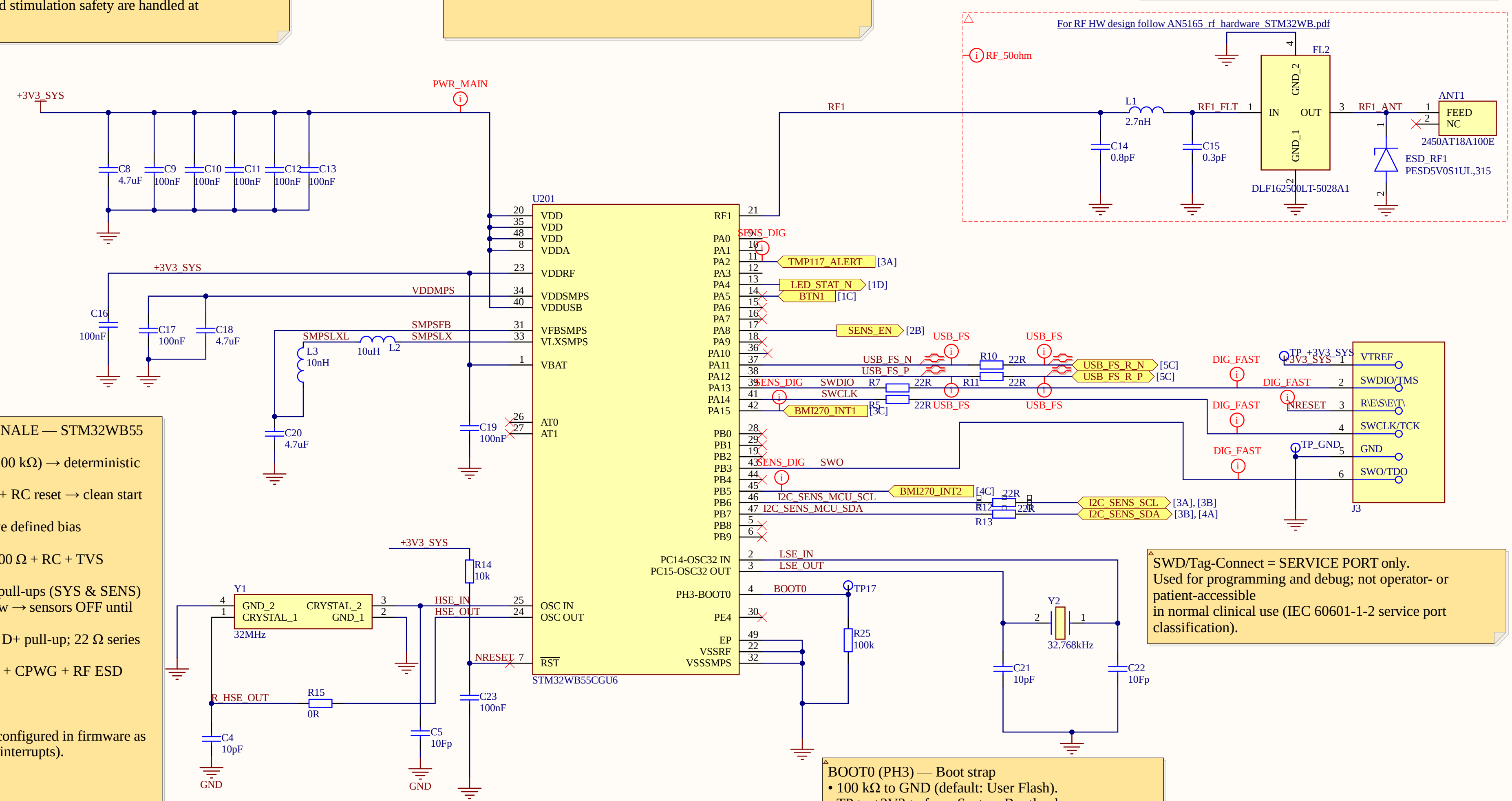
MCU (STM32WB55), SWD Debug & RF (BLE) — BLEC-SCH-0002

Essential performance (controller role):
- Board provides BLE command/control only; no direct patient-applied parts.
- All MCU I/Os are SELV and default to safe states on reset (pull-ups/-downs and series resistors as shown).
- BLE link integrity and stimulation safety are handled at system/implant level.

IEC 60601 Compliance:
USB-C = AC/DC SELV Input Port.
PPTC + TVS + ESD + CMC provide protection for IEC 61000-4-2/4-4/4-5.
All voltages SELV; battery safety provided by charger + pack.

RF port (intentional radiator):
2.4 GHz BLE antenna with π-match (C-L-C) and RF ESD footprint.
CPWG, via fence and π-match used to meet radiated emission/immunity
requirements (IEC 60601-1-2).



MCU DESIGN RATIONALE — STM32WB55

- BOOT0 pulled low (100 kΩ) → deterministic User Flash boot
- NRST has 10 kΩ PU + RC reset → clean start under EMC
- All interrupt lines have defined bias (TMP117/BMI)
- BTN1 has pull-up + 100 Ω + RC + TVS (false-trigger resistant)
- I²C buses use 4.7 kΩ pull-ups (SYS & SENS)
- SENS_EN defaults low → sensors OFF until MCU enables
- USB FS uses internal D+ pull-up; 22 Ω series resistors
- RF path uses π-match + CPWG + RF ESD footprint

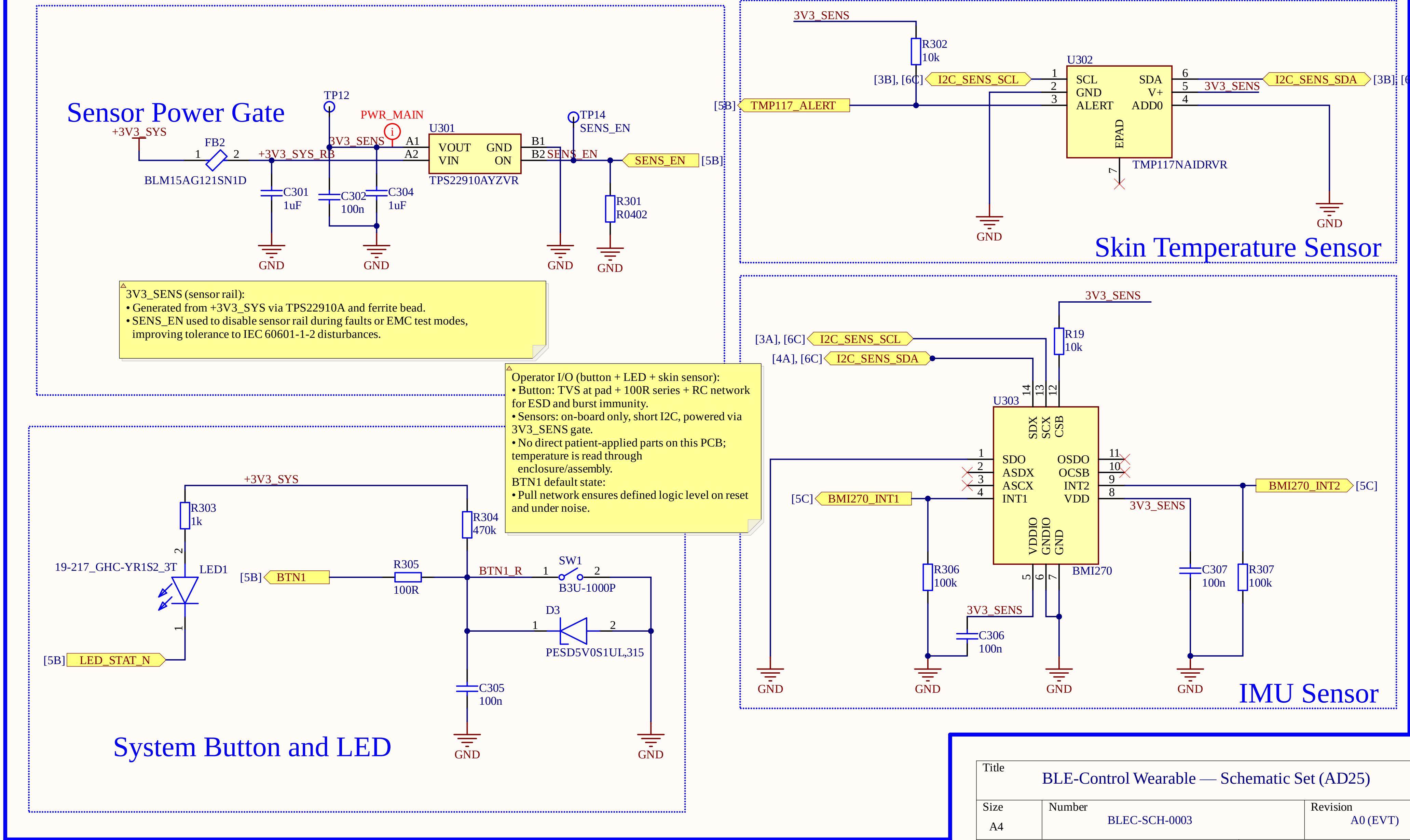
Unused GPIO Policy:
All unused MCU pins configured in firmware as ANALOG (no pull, no interrupts).

SWD/Tag-Connect = SERVICE PORT only.
Used for programming and debug; not operator- or patient-accessible
in normal clinical use (IEC 60601-1-2 service port classification).

BOOT0 (PH3) — Boot strap
- 100 kΩ to GND (default: User Flash).
- TP to +3V3 to force System Bootloader.
- Do not use as GPIO.

Title	BLE-Control Wearable — Schematic Set (AD25)	
Size: A3	Number: BLEC-SCH-0002	Revision: A0 (EVT)
Date: 5/26/2026	Sheet of: 2 of 3	
File: C:\Users\..\MCU_RF.SchDoc	Drawn By: Caoilte Donohoe	

Sensors/IO: TPS22910A 3V3_SENS, Button, LEDs, I²C — BLEC-SCH-0003



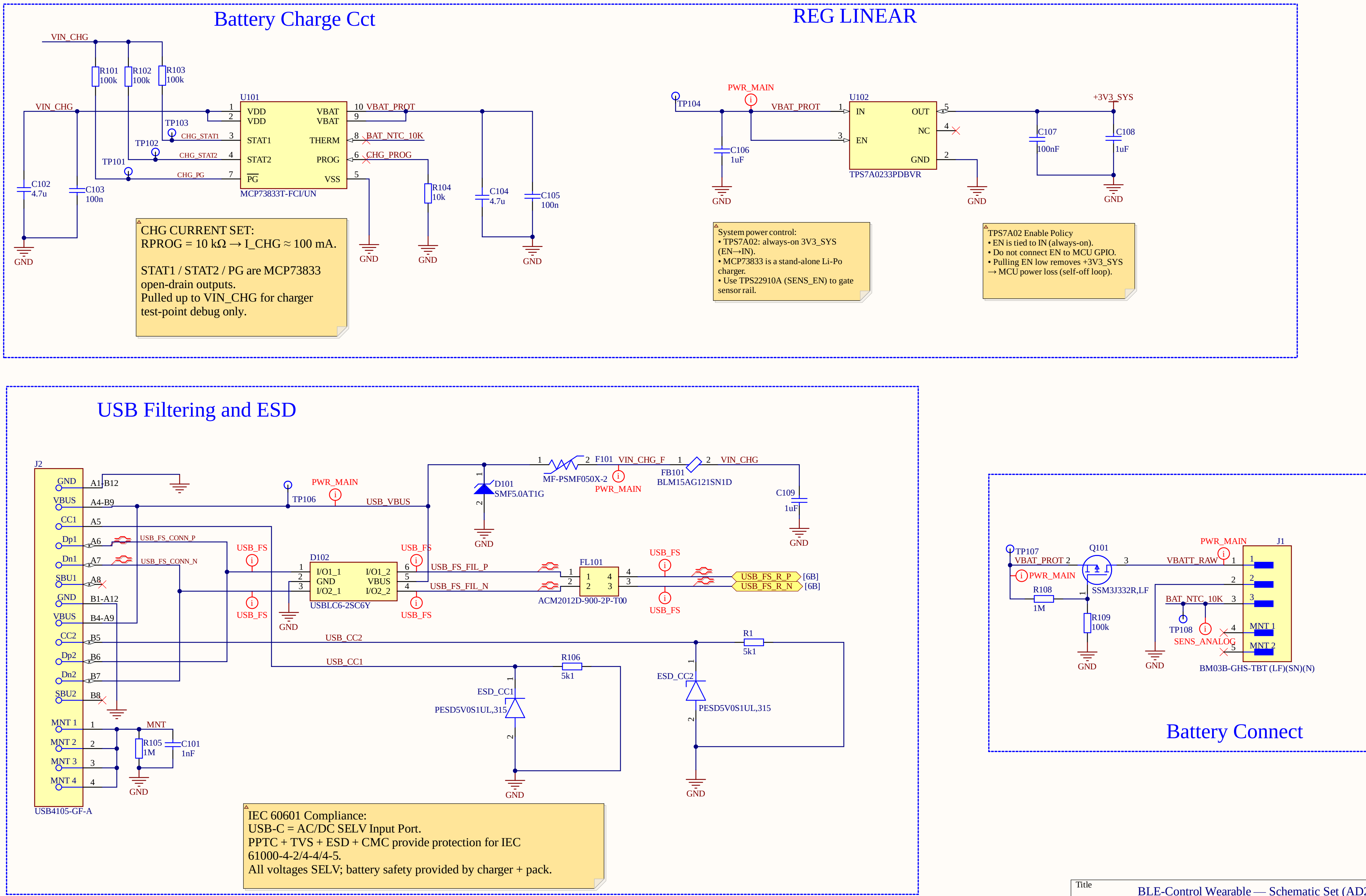
3V3_SENS (sensor rail):
- Generated from +3V3_SYS via TPS22910A and ferrite bead.
- SENS_EN used to disable sensor rail during faults or EMC test modes, improving tolerance to IEC 60601-1-2 disturbances.

Operator I/O (button + LED + skin sensor):
- Button: TVS at pad + 100R series + RC network for ESD and burst immunity.
- Sensors: on-board only, short I2C, powered via 3V3_SENS gate.
- No direct patient-applied parts on this PCB; temperature is read through enclosure/assembly.

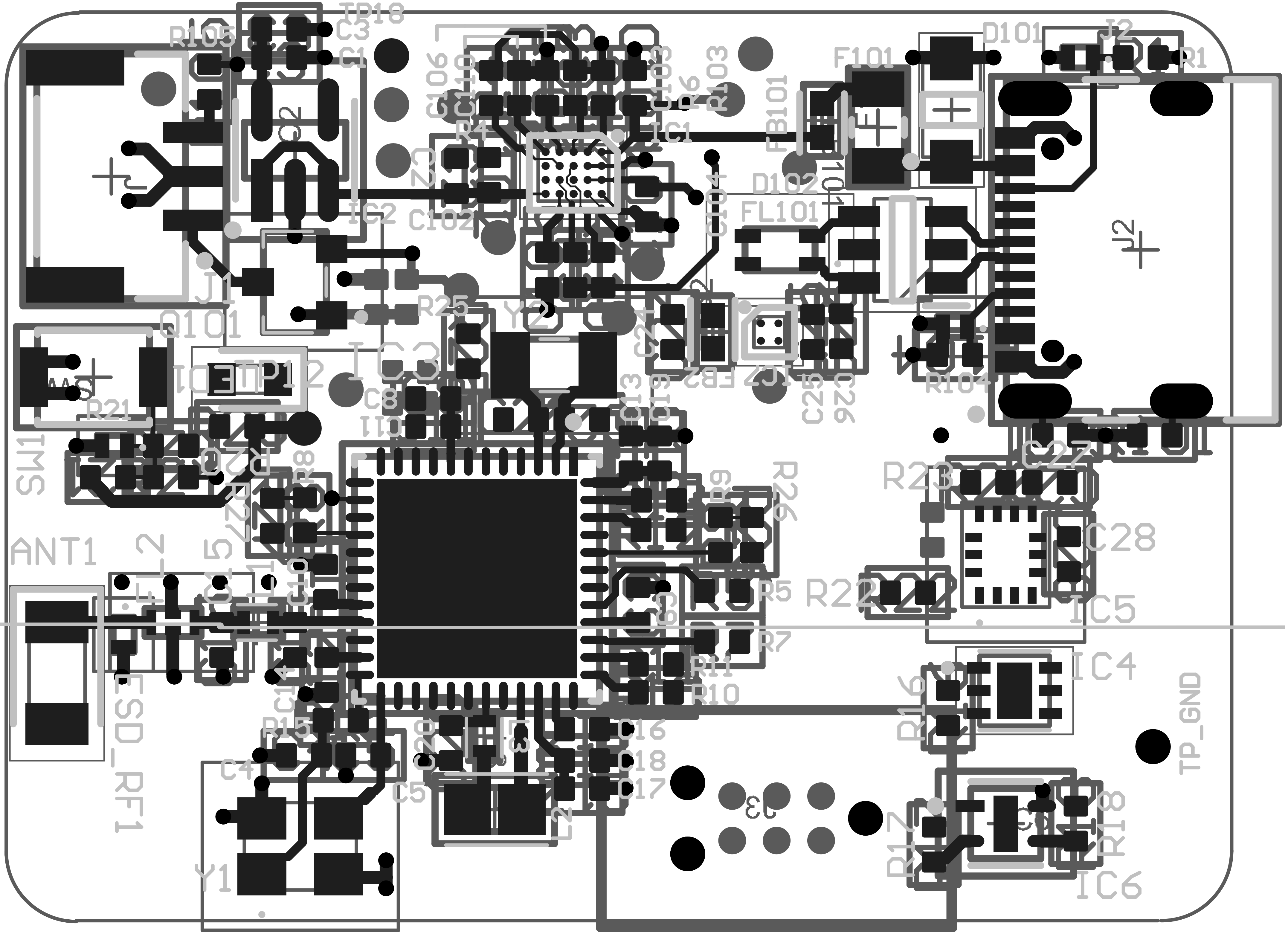
BTN1 default state:
- Pull network ensures defined logic level on reset and under noise.

Title	BLE-Control Wearable — Schematic Set (AD25)	
Size: A4	Number: BLEC-SCH-0003	Revision: A0 (EVT)
Date: 5/26/2026		Sheet of: 3 of 3
File: C:\Users\..\Sensor_IO_Buttons_LED.SchDoc		Drawn By: Caoilte Donohoe

Power, Battery, USB-C & MCP73833T (Main 3V3) — BLEC-SCH-0001



Title	BLE-Control Wearable — Schematic Set (AD25)		
Size: A3	Number: BLEC-SCH-0001		Revision: A0 (EVT)
Date: 5/26/2026		Sheet of: 1 of 3	
File: C:\Users\..\Power_Charge_USB.SchDoc		Drawn By: Caoilte Donohoe	


R105
TP18
C3
C1
C2
J1
IC2
C106
C110
RC1
C103
R6
R103
R4
IC1
FB101
F101
D101
J2
R1
Z3
C102
C104
D102
FL101
Q101
R25
Y2
IC3
TP12
C24
FB2
C25
C26
R104
SW1
R21
R20
R8
C8
C11
C13
C19
C27
R23
R9
R26
C28
IC5
R22
R5
R7
R11
R10
ANT1
ESD_RF1
C15
C14
R15
C4
C5
C20
L3
C16
C18
C17
L2
Y1
J3
IC4
R16
TP_GND
R17
R18
IC6